LEVEL 1 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

MARK SCHEME

Sample Assessment

Paper: RFSML1SAM01

Functional Skills in Mathematics Level 1 – Mark scheme

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Task 1 NC	Process (Task description)	Total mark	Mark allocation	Comments	PS or US	Subject content
Question 1	Calculate perimeter of shape	2	**1 mark**: Any valid method used to calculate perimeter, eg 7.4 + 12.6 + 11.4 + 6.2 + 4 + 6.4 OR (11.4 + 12.6) × 2	Units not required. Accept any other valid method. Accept if 48 seen.	US	22b
			1 mark: Correct perimeter shown ie 48m	Units not required.	US	22b
Question 2	Calculate square of 17	1	**1 mark**: (17 x 17) = 289		US	6
Question 3	Calculate number of marbles	1	**1 mark**: Correct number of white marbles: 7		US	17a
Question 4	Correct addition of numbers	1	**1 mark**: Correct answer 33.22		US	11a
Question 5	Correct division by 100	1	**1 mark**: Correct answer 0.468		US	3b

Question 6a	Calculate 3 sides of the garden area Correct method to find number of strips Correct number of border strips needed Cost found using estimate of numbers	4	**1 mark**: Correctly calculated 3 sides of the garden area (3 sides) = 42 (m)	Accept 42 seen.	PS	5
			1 mark: Correct method used following rule ie, 42 ÷ 3 × 2	FT from their calculation of 3 sides.	PS	5
			1 mark: Correct answer 28		PS	5
			1 mark: Valid method used to estimate, eg (10 × 30) = 300) OR (10 × 28) (= 280)	Allow FT for their number of border strips. Correct money notation not required. Do not award if 9.89 not rounded.	PS	12a
Question 6b	Conversion from ml to l or l to ml Calculate number fence panels	2	**1 mark**: Conversion from l to ml or ml to l, eg 1.5 × 1000 = 1500 OR 3 × 1000 = 3000 OR 300 ÷ 1000 = 0.3	Units not required. Award mark if 10 seen as their answer.	PS	20c
			1 mark: Correct number of fence panels, ie 10 panels		PS	20c
Question 6c	Valid method to calculate length or width Correct actual length and width shown	2	**1 mark**: Valid method to find appropriate length or width of table, eg 5.5 x 20 = (110 cm) OR 11 x 20 = (220 cm)	Units not required. May be implied if 110 or 220 seen.	PS	21
			1 mark: Correct length AND width of table shown, ie 110 (cm) and 220 (cm)	Both dimensions required for the mark. Units not required. Accept correct conversion to metres.	PS	21
Question 7	Identify missing dimension of the bedroom.	1	**1 mark**: 7.5 (m) identified 30 ÷ 4 = 7.5	Units not required. Award for correct answer seen.	PS	22a

Task 2	Process (Task description)	Total mark	Mark allocation	Comments	PS or US	Subject content (SoS)
Question 8	Express probability as a fraction	1	**1 mark**: 1/3 or one third shown	Accept 6/18	US	31
Question 9	Calculate percentage from fraction	1	**1 mark:** 62.5 (%)		US	16b
Question 10	Round to two decimals	1	**1 mark**: 6.67	Do not accept 6.66.	US	12b
Question 11a	Calculate amount of flour needed to make cakes.	4	**1 mark**: Valid method used to find amount of flour needed, eg 72 ÷ 12 (= 6) AND 6 × 400 OR 400 ÷ 12 (= 33.33) AND 33.33 × 72 OR 2399.99 or 2400 seen	May be implied if 350 or 0.350 seen for amount of flour left over.	PS	17b
	Convert fraction to decimal		**1 mark**: Conversion of ¾ kg to decimal, g or kg, eg 2.75kg OR 0.75kg OR 2750g OR 750g.	May be implied if 350 or 0.350 seen for amount of flour left over.	PS	16a
	Calculate amount of flour left over		**1 mark**: Correct amount of flour left over (based on rounded number of cakes), eg 2750 – 2400 = 350(g) OR 0.35 (kg).	Do not award for 150g or 0.15 kg. Allow FT for their amount of flour.	PS	20b
	Show correct units		**1 mark**: Correct units shown (g or kg) for their answer.	Allow FT for incorrect calculations. Do not allow 350kg or 0.35g.	PS	20b
Question 11b	Calculate time taken to prepare and bake loaves of bread Show amount of time taken	3	**1 mark**: Valid method used for adding up time taken, eg (6 × 7) + 45m + 10m (= 97m).	May be implied if 97 seen.	PS	20e
			1 mark: Correct time of 97 (minutes).	Units not required.	PS	20e
	Show time to start making loaves		**1 mark**: Correct time given to start making loaves of bread, eg 4.38 (am)	Allow FT from their calculated time.	PS	20e

Question 11c	Conversion from pence to pounds	4	**1 mark**: Evidence of conversion from pence to pounds or vice versa, eg 0.13 OR 0.56 OR 140 OR 2.60 OR 8.40 OR 13.80	Award if 13.8 seen.	PS	20d
	Method for calculating percentage		**1 mark:** Method to calculate percentage discount, 20 ÷ 100 x 13.80 OR 0.2 x 13.80 OR Other valid method	Award if 2.76 seen and FT	PS	19
	Calculate percentage discount		**1 mark**: Correct 20% discount, ie 2.76	Correct money notation not required.	PS	19
	Calculated discounted price		**1 mark**: Correctly calculated price after discount, ie 11.04	Correct money notation not required.	PS	19
Question 11d	Approximation of the trade discount	1	**1 mark**: Valid method to check the trade discount, eg 20 ÷ 100 x 14 OR 0.2 x 14	Accept any valid method to approximate answer.	PS	12a

<table>
<tr><th>Task 3</th><th>Process
(Task description)</th><th>Total mark</th><th>Mark allocation</th><th>Comments</th><th>PS or US</th><th>Subject content</th></tr>
<tr><td>Question 12</td><td>Write number in digits</td><td>1</td><td>1 mark: Correctly writing the number in digits, ie
190493</td><td>Award if comma or space between 1000s and 100s.</td><td>US</td><td>1a</td></tr>
<tr><td>Question 13</td><td>Identify highest number</td><td>1</td><td>1 mark: Bank E (4.76) identified.</td><td>Award for correct bank or interest rate identified.</td><td>US</td><td>10</td></tr>
<tr><td>Question 14</td><td>Complete frequency table</td><td>1</td><td>1 mark:
<table><tr><th>Number of marks</th><th>Frequency</th></tr><tr><td>0 – 9</td><td>0</td></tr><tr><td>10 – 19</td><td>2</td></tr><tr><td>20 – 29</td><td>4</td></tr><tr><td>30 – 39</td><td>6</td></tr><tr><td>40 – 49</td><td>4</td></tr></table></td><td>Allow tally or totals.</td><td>US</td><td>28a</td></tr>
<tr><td rowspan="2">Question 15a</td><td>Correct stat shown for matches</td><td rowspan="2">3</td><td>2 marks: Correct values shown for all matches, ie
Match 1: 3
Match 2: 3
Match 3: 4
Match 4: -2</td><td>Award 1 mark for any 2 correct values shown.</td><td>PS</td><td>2</td></tr>
<tr><td>Correct totals</td><td>1 mark: Correct values shown for totals row, ie
Totals: 7, 13, -6</td><td></td><td>PS</td><td>2</td></tr>
<tr><td>Question 15b</td><td>Explain probability</td><td>1</td><td>1 mark: Correct answer with explanation, eg
No, because there is 0.88 chance of rainfall, which means there is a high chance that it will rain on Saturday OR
Other valid explanation</td><td>Do not accept ‘no’ without explanation.</td><td>PS</td><td>30</td></tr>
<tr><td rowspan="3">Question 15c</td><td>Valid method to find perimeter of pitch</td><td rowspan="3">3</td><td>1 mark: Valid method to find perimeter of pitch, eg
18 + 18 + 36 + 36 = OR
(18 × 2) + (36 × 2) = OR
Any other correct method</td><td></td><td>PS</td><td>22b</td></tr>
<tr><td>Conversion from m to km or km to m</td><td>1 mark: Evidence of conversion from m to km or vice versa. Eg
0.108 OR
1000m</td><td>Units not required</td><td>PS</td><td>20a</td></tr>
<tr><td>Correct number of laps</td><td>1 mark: Correct number of full laps around the pitch, ie
10</td><td>Do not accept 9 laps/times around the pitch</td><td>PS</td><td>12a</td></tr>
</table>

Question 15d	Calculate percentage	3	**1 mark:** correct method to calculate percentage**,** eg 35 ÷ 100 x 380 OR 0.55 x 380 OR 20 ÷ 100 x 380 OR 0.2 x 380 OR 0.45 x 380 OR 45 ÷ 100 x 380 OR Other valid percentage calculation	May be implied if 209, 133 or 76 seen. Award if 171 seen.	PS	14
			1 mark: correct number of adult tickets, eg 171 adult tickets sold		PS	14
			1 mark: correct answer, eg 'No, Ryan was not correct'	Only award if valid calculation AND/OR 171 seen	PS	14
Question 15e	Subtract decimals from decimals Calculate answer	2	**1 mark:** correct subtraction method, eg (2.94 x 380 =) 1117.2 AND 3697.40 – 1117.2	Award if 2580.20 seen. FT for incorrect total donation.	PS	11b
			1 mark: correct answer, eg £2580.20	£ sign not required.	PS	11b

Task 4	Process (Task description)	Total mark	Mark allocation	Comments	PS or US	Subject content
Question 16	Appropriate scale given. Bars at correct heights Graph appropriately labelled	3	**1 mark**: appropriate scale given	Do not award for line graph.	US	27b
			1 mark: bars at correct height (tolerance plus/minus 1 division)		US	27b
			1 mark: Graph contains appropriate axis labels and title, eg X axis: Months Y axis: Laptops Title: Graph to show number of laptops sold over 6 months	Accept similar wording for axis labels and title.	US	27b
Question 17a	Identify correct net Justify answer	2	**1 mark**: Net A.	Do not award without supporting valid explanation.	PS	25b
			1 mark: Any valid reason, eg "Net B is the shape of a cube so does not match the picture." OR "The other two boxes are too high compared to the picture." OR "The height of the box in the picture is very small which matches the dimensions of Net A." OR "Net C does not have a lid" OR "Net D dimensions are too large"	Accept any valid reason given for choosing their net.	PS	25b
Question 17b	Calculate number of small boxes that will fit in large box	4	**1 mark**: Valid method used to calculate number of small boxes that will fit in **either** large box, eg **Box A method** 50 ÷ 10 = 5 8 ÷ 8 = 1 12 ÷ 3 = 4 AND 5 × 1 × 4 (= 20) OR **Box B method** 50 ÷ 10 = 5 16 ÷ 8 = 2 15 ÷ 3 = 5 AND 5 × 5 × 2 (= 50) OR **Box A (volume method)** 10 × 8 × 3 = 240 50 × 8 × 12 = 4800 4800 ÷ 240 (= 20) OR **Box B (volume method)** 50 × 15 × 16 = 12000		PS	23

	Identify correct number of small boxes that will fit in large box Calculate number of large boxes needed for 100 bracelets Calculate cost of buying enough large boxes		12000 ÷ 240 (= 50)			
			1 mark: Correct answer given for **either** box. Box A: 20 OR Box B: 50		PS	23
			1 mark: Correct number found for **both** boxes, eg Box A: 100 ÷ 20 = 5 Box B: 100 ÷ 50 = 2	Allow FT for their number of small boxes per large box providing answer is feasible.	PS	23
			1 mark: Correct calculation and answer given for cost of each box, eg Box A: 5 × £0.70 = £3.50 Box B: 2 × £1.80 = £3.60	Allow FT for their number of boxes calculated.	PS	23
Question 18a	Calculate mean of bracelets sold, or totals of necklaces, rings and earrings	2	**1 mark**: Correct mean number of bracelets sold, eg 22 + 28 + 23 + 38 + 44 + 97 = 252 AND 252 ÷ 6 = 42 OR Correct total of either necklaces, rings and earrings sold, eg 47 × 6 = 282 necklaces OR 35 × 6 = 210 rings OR 39 × 6 = 234 earrings	Award if 42 seen Award if 282 or 210 or 234 seen	PS	29a
	Identify bestselling item		**1 mark**: Necklace identified as bestselling item.	Do **not** allow FT for incorrect calculations. Do not award if not supported by calculations	PS	29a
Question 18b	Calculate range of bracelets sold	2	**1 mark**: Correct range calculated, eg 97 – 22 = 75 identified (maximum and minimum identified).		PS	29b
	Identify most consistent item		**1 mark**: Rings identified as most consistent selling item.	Do not allow FT for incorrect calculations.	PS	29b
Question 18c	Calculate fraction of amounts	2	**1 mark:** Method to calculate fraction of amounts eg 1592 ÷ 3 x 2 = (1,061.33…) OR 1 ÷ 3 x 1592 = (530.66…)		PS	9
			1 mark: Correct answer = (£)1061.33	Allow 1061.34 Only allow 2 decimal places.	PS	9

Annotation notes:

Annotation	Meaning
US	Underpinning skills
PS	Problem solving skills
FT	Follow through
(…)	Information that is not required for the mark point